13.5.3. Structure of benzene:

1. Molecular formula

Elemental Analysis and molecular weight determination have proved that the molecular formula of benzene is C_6H_6. This indicates that benzene is a highly unsaturated compound.

2. Straight chain structure not possible:

Benzene could be constructed as a straight chain or ring compound but it not feasible since it does not show the properties of alkenes or alkynes.for example, it did not decolourise bromine in carbon tetrachloride or acidified $KMnO_4$. It did not react with water in the presence of acid.

3. Evidence of cyclic structure:

I) substitution of benzene:

Benzene reacts with bromine in the presence of $AlCl_3$ to form mono bromo benzene.

$$C_6H_6 + 3Br_2 \xrightarrow{AlCl_3} C_6H_5Br + HBr$$

bromobenzene

Formation of only one monobromo compound indicates that all the six hydrogen atoms in benzene were identical. This is possible only if it has a cyclic structure of six carbons each containing one hydrogen.

II) addition of hydrogen:

Benzene can add on to three moles of hydrogen in the presence of nickel catalyst to give cyclohexane.

$$C_6H_6 + 3H_2 \xrightarrow{\text{Raney Ni}} C_6H_{12}$$

cyclohexane

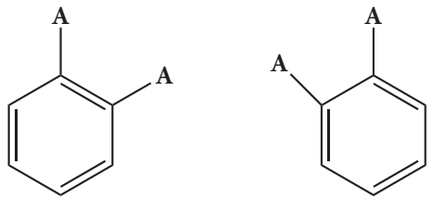
This confirms cyclic structure of benzene and the presence of three carbon-carbon double bond.

4. Kekule's structure of benzene:

In 1865, August Kekule suggested that benzene consists of a cyclic planar structure of six carbon with alternate single and double bonds.

There were two objections:

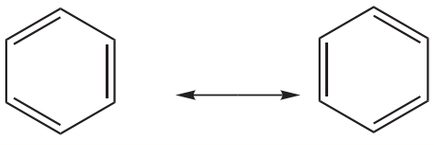
(i) Benzene forms only one ortho disubstituted products whereas the Kekule's structure predicts two o-di substituted products as shown below.



Presence of double bond between the substituents

Presence of single bond between the substituents

(ii) Kekule's structure failed to explain why benzene with three double bonds did not give addition reactions like other alkenes. To overcome this objection, Kekule suggested that benzene was mixture of two forms (1 and 2)which are in rapid equilibrium.



5. Resonance description of benzene:

The phenomenon in which two or more structures can be written for a substance which has identical position

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